

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0735

Stephensburg 138 kV
September 27, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0735. The Interconnection Customer proposes a 361.7 MW natural gas combined cycle facility interconnecting at the Stephensburg 138 kV substation in the Pennsylvania area. The request was received on January 28, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 267 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$25,771,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Stephensburg-Hosensack 138 KV Line	Omak-Stephensburg Substation 138 KV	185.4	183	100.0
Omak-Stephensburg 138 KV	Stephensburg-Hosensack 138 KV	155.6	218	100.0
Chico B-Hosensack 138 KV	Omak-Stephensburg 138 KV line	150.1	209	100.0
Chico B-Alburtis 138 KV	Omak-Stephensburg 138 KV	138.3	220	100.0
Omak-Berrenda C Substation 138 KV	Stephensburg-Hosensack 138 KV	135.8	235	100.0
Cornelius Delivery No 1-Alburtis 138 KV	Stephensburg-Omak 138 KV	123.7	233	100.0
Amistad-Walpole 138 KV Line	Arbuckle-Berrenda C 345 KV	112.9	243	29.2
Walpole-Berrenda C 138 KV	Arbuckle-Berrenda C 345 KV L	104.2	271	24.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Stephensburg-Hosensack 138 KV Line	Rebuild line with higher-capacity conductor	\$3,553,000	100% (\$3,553,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Omak-Stephensburg 138 KV	Rebuild line with higher-capacity conductor	\$4,841,000	100% (\$4,841,000)
Chico B-Hosensack 138 KV line	Rebuild line with higher-capacity conductor	\$4,900,000	100% (\$4,900,000)
Chico B-Alburtis 138 KV	Rebuild line with higher-capacity conductor	\$4,231,000	100% (\$4,231,000)
Omak-Berrenda C Substation 138 KV	Rebuild line with higher-capacity conductor	\$1,633,000	100% (\$1,633,000)
Cornelius Delivery No 1-Alburtis 138 KV	Rebuild line with higher-capacity conductor	\$1,732,000	100% (\$1,732,000)
Amistad-Walpole 138 KV Line	Rebuild line with higher-capacity conductor	\$2,398,000	100% (\$2,398,000)
Walpole-Berrenda C 138 KV	Reconductor line	\$2,483,000	100% (\$2,483,000)
	Total allocated to project		\$25,771,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.